

Module 14: Bridges, Switches, Spanning Tree Protocol (STP) & Unit 2 AKTU PYQs

Unit 2: Data Link Layer & MAC Sublayer

Layer 2 Switching & AKTU Exam Archive

Deep-dive into transparent bridge MAC address learning, filtering, and flooding, bridging loops, IEEE 802.1D Spanning Tree Protocol root election, and verified model solutions for AKTU BCS603 exam questions.

Transparent Learning Bridges & Spanning Tree Protocol (IEEE 802.1D STP)

1. Transparent Bridge Learning Algorithm



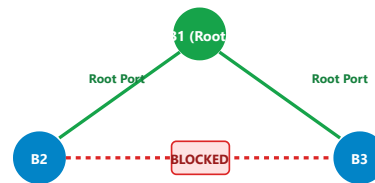
• The Three Forwarding Rules:

1. **LEARN:** Record Source MAC + Arrival Port in Forwarding Database.
2. **FORWARD / FILTER:** If Dest MAC is known on another port → Send ONLY there. If Dest MAC is on the SAME port as source → FILTER (Drop frame, save link!).
3. **FLOOD:** If Dest MAC is unknown or Broadcast → Forward to ALL ports except arrival!

Collision Domain vs Broadcast Domain:

Bridge/Switch breaks Collision Domains into 1 per port. Maintains 1 single Broadcast Domain.

2. Spanning Tree Protocol (STP 802.1D)



• The 4-Step STP Algorithm:

1. **Elect Root Bridge:** Bridge with Lowest Bridge ID (Priority + MAC).
2. **Select Root Ports (RP):** Port on each non-root bridge with lowest path cost to Root.
3. **Select Designated Ports (DP):** Port per LAN segment with lowest path cost to Root.
4. **Block Alternate Ports:** Put remaining loop-causing ports in **BLOCKING** state!

AKTU BCS603 Unit 2 Core Previous Year Exam Questions (10 Marks Distribution)

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| Q1 (Framing & Stuffing): | Explain Bit Stuffing (5 ones rule) and Byte Stuffing (ESC character). Given bit stream, apply bit stuffing. |
| Q2 (ALOHA & CSMA/CD): | Derive Throughput of Pure and Slotted ALOHA. Prove $T_t \geq 2T_p$ condition in CSMA/CD and 64-byte frame size. |
| Q3 (Flow Control ARQ): | Derive Stop-and-Wait efficiency $\eta = 1/(1+2a)$. Prove why $W_s = 2^m - 1$ in GBN and $W_s = 2^{m-1}$ in Selective Repeat. |
| Q4 (Error & STP): | Solved CRC numerical with polynomial $G(x)$, Hamming (7, 4) syndrome calculation, and STP root election algorithm. |

1. Transparent Bridge Operation

Layer 2 switches partition collision domains on a per-port basis while preserving a single broadcast domain. Forwarding tables dynamically learn source MAC addresses and intelligently filter or forward frames, eliminating unnecessary collisions.

2. Spanning Tree Protocol (STP 802.1D)

STP Algorithmic Steps

1. Elect Root Bridge (Lowest Bridge ID = Priority + MAC)
2. Select Root Port on non-root bridges (Lowest path cost to Root)
3. Select Designated Port per segment (Lowest path cost to Root)
4. Place remaining redundant loop-causing ports in BLOCKING state